

Homework 3: SQL

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Problem 1:

Given the following relational database:

employee (employee-name, street, city)

works (employee-name, company-name, salary)

company (company-name, city)

manages (employee-name, manager-name)

Write the following Queries in SQL:

1. Find the names of employees who work in a city different from the city where their company is located.

```
SELECT e.employee-name
FROM employee e, works w, company c
WHERE e.employee-name = w.employee-name
AND w.company-name = c.company-name
AND e.city <> c.city; -- A "not equal" comparison operator
```

2. Find the name of employees who earn less than the average salary of their company.

```
SELECT e.employee-name
FROM employee e, works w
WHERE e.employee-name = w.employee-name
AND w.salary < (
    SELECT AVG(salary)
    FROM works
    WHERE company-name = w.company-name
)
```

Problem 2: Write the following queries in SQL on database shown below.

EMPLOYEE								
FNAME	LNAME	<u>SSN</u>	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNUM

DEPARTMENT			
DNAME	<u>DNUMBER</u>	MGRSSN	MGRSTARTDATE

DEPT_LOCATIONS	
<u>DNUMBER</u>	<u>DLOCATION</u>

WORKS_ON		
<u>ESSN</u>	<u>PNUM</u>	HOURS

PROJECT			
PNAME	<u>PNUMBER</u>	PLOCATION	DNUM

DEPENDENT				
<u>ESSN</u>	<u>DEPENDENT_NAME</u>	SEX	BDATE	RELATIONSHIP

Figure 1: Database Schema for Problem 2

EMPLOYEE (FNAME, LNAME, SSN, BDATE, ADDRESS, SEX, SALARY, SUPERSSN, DNUM)

DEPARTMENT (DNAME, DNUMBER, MGRSSN, MGRSSTARTDATE)

DEPT_LOCATIONS (DNUMBER, DLOCATION)

WORKS_ON (ESSN, PNO, HOURS)

PROJECT (PNAME, PNUMBER, PLOCATION, DNUM)

DEPENDENT(ESSN, DEPENDENT_NAME, SEX, BDATE, RELATIONSHIP)

1. For departments with an average salary greater than \$50,000, list the department name and the average salary of employees in each department.

```
SELECT d.dname, AVG(w.salary) AS average_salary
FROM department d, employee e
WHERE d.DNUMBER = e.DNUM
GROUP BY d.dname
HAVING AVG(w.salary) > 50000
```

2. Find the first and last names of employees who do not work on any project.

```
SELECT e.fname, e.lname
FROM employee e
WHERE NOT EXISTS (SELECT * FROM works_on w WHERE e.SSN = w.ESSN)
```

3. Retrieve the last names of employees who share a birthday with at least one of their dependents.

```
SELECT DISTINCT e.lname
FROM employee e, dependent d
WHERE e.SSN = d.ESSN
AND e.BDATE = d.BDATE
```

4. Find the first and last names of employees who work on every project in the “Houston” location.

```
SELECT e.fname, e.lname
FROM employee e
WHERE NOT EXISTS (
  -- Find a Houston Project
  SELECT *
  FROM project p
  WHERE p.PLOCATION = 'Houston'
  AND NOT EXISTS (
    -- That the current employee does NOT work on
    SELECT *
    FROM works_on w
    WHERE w.ESSN = e.SSN
    AND w.PNO = p.PNUMBER
  )
)
```

5. For each department, list the department name and the number of employees who work in that department.

```
SELECT d.dname, COUNT(e.SSN) AS num_employees
FROM department d
LEFT JOIN employee e ON d.DNUMBER = e.DNUM
GROUP BY d.dnumber, d.dname;
```

6. List the last names of all employees who have no dependents

```
-- No need for DISTINCT because E with no D only appear once
SELECT e.lname
FROM employee e
WHERE NOT EXISTS (
  SELECT *
  FROM dependent d
  WHERE e.SSN = d.ESSN
)
```

Problem 3: Consider the following relational university database:

STUDENT (Name, StudentNumber, Class, Major)

COURSE (CourseName, CourseNumber, CreditHours, Department)

PREREQUISITE (CourseNumber, PrerequisiteNumber)

SECTION (SectionIdentifier, CourseNumber, Semester, Year, Instructor)

GRADE_REPORT (StudentNumber, SectionIdentifier, Grade)

1. Retrieve the names and majors of all 'A' students (i.e., students who have a grade of 'A' in all their courses).

```
SELECT s.Name, s.Major
FROM STUDENT s
WHERE NOT EXISTS (
    SELECT *
    FROM GRADE_REPORT g
    WHERE g.StudentNumber = s.StudentNumber
    AND g.Grade <> 'A'
)
-- Making sure they at least have one grade on record
AND EXISTS (
    SELECT *
    FROM GRADE_REPORT g2
    WHERE g2.StudentNumber = s.StudentNumber
)
```

2. Retrieve the names and majors of all students who do not have a grade of 'A' in any of their courses

```
SELECT s.Name, s.Major
FROM STUDENT s
-- Removes all students with an 'A' grade in any course
WHERE NOT EXISTS (
    SELECT *
    FROM GRADE_REPORT g
    WHERE g.StudentNumber = s.StudentNumber
    AND g.Grade = 'A'
)
```